Trace-zero leads: restrict and certify before solving

2026-11-09 · Isogeny-labs/crypto · PR3744 · local unsigned evidence

Result: Two concrete specializations are implemented and checked. A finite admitted domain can replace redundant full-point membership witnesses. On the actual small polynomial window, restricting q(x)=Tr(xH(x)) before coordinate elimination preserves quadratic degree. Two further lead cards specify complete quadratic relation chains and rank-aware presentation choice.

Two checked specializations; two next constructions

E_0 / F_2 · ECC2K-130 uses n=131 · local evidence, unsigned · 2026-11-09

TZL-01 · Certified finite domain

Validate actual T_n points → retain exact x domain → eliminate redundant y witnesses

All identity patterns and actual-base lifting remain; final original-group replay is mandatory.

Constructed SAT variables · n131, m3, all-affine pattern, one allowed G abscissa



Counts from exact circuit construction; no solver timing, natural relation yield or scaling claim.

TZL-02 · Restrict q(x)=Tr(xH(x)) to the existing V first

V=span(z,…,z^{16}): q|V=0. Full membership gives Tr(y)=0 without a cubic substitution.

Same point set: raw 147 variables / 132 rows → compact 146 / 131; both degree 2.

Counter-control: n83,d64 has polar rank48; compact is cubic. No universal degree reduction.

TZL-03 · Proposed

Complete quadratic full-point chains

First gate: exact branch/lift completeness.

TZL-04 · Proposed

Rank-aware matched presentation selection

First gate: charged held-out successful work.

Source: trace_zero_leads_20261109/models.json and checks/new-tests.txt · one constructed instance per circuit row

TZL-01: finite-domain membership certificate

Every allowed x is taken from an original point admitted to the full group-trace kernel T_n. Its negation is also in T_n. The added ordinate witnesses occur only inside each summand's membership model, so every allowed x has a satisfying choice independently of the S3 constraints. Existentially removing those witnesses preserves the projected system. Actual supplied base signs still matter and are checked during lifting and original-group replay.

The new API decompose_full_trace_zero_with_encoding accepts CertifiedDomain as an opt-in alternative. Default ExplicitPoints is retained. Point admission, finite x domains, identity patterns, original-group acceptance and conservative resource caps remain. This specialization does not apply to an arbitrary unadmitted algebraic subspace. Counts above concern the same singleton-domain instance, before full-point pins; they are not timing evidence.

TZL-02: the restricted second bit can be linear

For odd n, the half-trace identity gives q(x)=Tr(x)+St(x), where St is the second elementary symmetric polynomial of all Frobenius conjugates of x. The H term at exponent0 gives $Tr(x^2)=Tr(x)$. Its remaining even exponents choose one from each nonzero difference class modulo n up to sign, so their traces sum every unordered conjugate pair once.

Polarization therefore gives Bq(x,z)=Tr(xz)+Tr(x)Tr(z). On V inside the field-trace kernel, Bq is simply the trace pairing. If its restriction is zero, q on V is linear, but need not vanish. Existing production halving then makes full affine membership Tr(y)=q(x).

The actual ECC2K-130 window V=span(z,…,z^{16}) has q restricted identically zero. It is the same retained subspace from the original span(1,z,…,z^{16}) after its x-trace coordinate is removed. No new factor base is selected. Its product-span dimension is31. Restricting before eliminating y_0 gives:

n131,d16	Variables	Rows	Terms	Degree	AND aux
Raw	147	132	3108	2	2216
Compact	146	131	3076	2	2200
Factored	147	132	3094	2	2216

All three rows describe the same allowed affine points. Boolean variables are implicit, with a separate identity branch. These are exact finite-point models, not general geometric defining ideals. The earlier unrestricted 260-variable model describes another point set and is not this comparison’s baseline.

n	d	Polar rank	Linear q terms	Quadratic q terms	dim(V·V)
83	16	0	0	0	31
83	32	0	1	0	63
83	64	48	4	56	83
131	16	0	0	0	31
131	32	0	0	0	63
131	64	0	1	0	127

These are constructed profiles, one exact basis per row. At n131,d64 the remaining q bit is the z^{59} coordinate, so a zero-polar-form assertion must not be promoted to q=0. At n83,d64 compact becomes cubic and has10646 AND auxiliaries against7328 for raw/factored. This counter-control is retained.

Integral certificate IDC1h3f66989fcfd56337 records the polarization expansion of S2 in16 formal coordinates, degree2,32 issued plus32 local replay points over$2^{61}-1$. It concerns that integral identity only, not the characteristic-two descent or a formal theorem. The written expansion and production-source native checks carry the field-model argument; independent standing is unverified.

New deciding steps

TZL-01: compare explicit/certified modes on identical non-singleton bases, unseen known-answer targets, seeds, all identity patterns and total budgets. Charge admission, rejected models and successful replay. Reduced allocation could also weaken propagation; only complete successful work settles it.

TZL-02: instantiate matched relation systems using restricted raw, compact and factored point models. Record actual solving degree and complete work on the same points and targets. Input degree alone is not solving degree.

TZL-03, proposed: retain full points and slope witnesses to obtain quadratic addition constraints instead of eliminating y into S3. Include generic addition with nonzero guards, doubling, opposite points and identity branches. Do not constrain partial sums to the original V. First compare every small-field case with the actual group. At m3,d16,n131 the summand models alone have438 variables, against179 in the naive x-only chain layout. The extra-variable tradeoff must be charged; no complete builder or solver result exists yet. S3 also admits quadratic lifting, already used by the native AND/XOR circuit. Compare that baseline too; displayed input degree alone supplies no advantage.

TZL-04, proposed: use q polar rank, product-space dimension and actual term/ auxiliary counts to select a presentation while keeping V fixed. Test held-out successful work against every fixed mode and a charged portfolio. Profile statistics are not a solver-cost theorem; changing the subspace is a separate controlled experiment.

Validation and current limits

Seven new focused tests pass, including exhaustive restricted Boolean models at n3/5/7, exact relations with sign-restricted and multiple-x finite bases, malformed/cap rejection, independent small-field and public n83/131 subtrace checks, and certificate replay/tamper rejection. The old ten-test full-variety suite also passes. Both harnesses use actual production modules without arithmetic replacements. Root cargo check –lib –no-default-features passes.

Certified singleton-domain identity controls are SAT at n83 and131 with415 and655 SAT variables, no ordinate pins, and original G+(-G)+G=G replay. The original explicit n83 control still returns UNKNOWN at the same one-second/ 2000-conflict cap. These are correctness controls, not natural-target relations or a general performance comparison. The earlier whole-root failures and benchrec limitation remain on the PR. No EKS or paid job ran; checkpoint/S3 access is still missing for matched experiments.

Flow: prior-art review, two implementations, exact constructions and local checks completed; TZL-03/04 proposed; natural relation-finding, performance and independent signed validation not attempted. Existing scoreboard and performance graph values remain unchanged.

Sources and reproduction

README.md gives locked commands and proof scope. LEADS.json maps all four cards to existing atlas families. models.json preserves exact bases, restricted ANFs and circuit counts; checks/ retains raw output and environment. Source hashes pin this revision; previous studies retain their historical identities.

Sources: current crypto-autoresearcher ECDLP atlas, decomposition-encoding-and-sat, factor-base-subspace-choice, cofactor-torsion-quotients and prime-degree-algebraic- decomposition rows; Kosters–Yeo1503.08001; Trimoska–Ionica–Dequen AFRICACRYPT2020; Ri–Myong–Kim–Rim1304.0521, definitions/Lemma3; existing production halving and addition derivations. KB unavailable (wrapper127); PMC follow-up recaptcha. No global novelty or new formally checked mathematical theorem is asserted.